📝 BUS 4101: E-Commerce Class Test (Architecture, Marketing & Logistics)

Focus: Classes 04–06 | JUST Dept. of CSE

1. Class 04: Architecture & Web Technologies

Three-Tier Architecture

- **Tier 1: Presentation Layer (Front-End):** What users see (HTML/CSS/JS, React).
- **Tier 2: Business Logic Layer (Application Server):** The "brain" (Python, Java, Node.js). Handles rules, orders, and auth.
- **Tier 3: Data Layer (Database):** Persistent storage (MySQL, MongoDB). Isolated for security.

REST vs. SOAP

Feature	REST	SOAP
Format	JSON (compact/fast)	XML (verbose/heavy)
Protocol	Architectural Style	Strict Protocol
Use Case	Public APIs, Mobile apps	Banking, Legacy systems

Scalability & Performance

- **Horizontal Scaling (Scale Out):** Adding more machines (e.g., Daraz 11.11 spike). Most cost-effective.
- **Vertical Scaling (Scale Up):** Adding power to one machine (RAM/CPU). Has physical limits.
- **CDN (Content Delivery Network):** Geographically distributed servers that cache static content (images/CSS) to reduce latency.

2. Class 05: Digital Marketing & Analytics

The AIDA Funnel

1. **Awareness:** SEO, Social Ads (Goal: Reach).
2. **Interest:** Blogs, Newsletters (Goal: Educate).

3. **Desire:** Reviews, Testimonials, Live Chat (Goal: Preference).
4. **Action:** Optimized Checkout, Trust Badges (Goal: Close Sale).

Key Formulas (High Exam Probability)

- Conversion Rate (CR): $CR = \frac{\text{Purchases}}{\text{Visits}} \times 100$
- Customer Acquisition Cost (CAC): $CAC = \frac{\text{Total Marketing Spend}}{\text{New Customers Acquired}}$
- Average Order Value (AOV): $AOV = \frac{\text{Total Revenue}}{\text{Number of Orders}}$
- Customer Lifetime Value (CLV): $CLV = AOV \times \text{Purchase Frequency} \times \text{Customer Lifespan}$
- Return on Ad Spend (ROAS): $ROAS = \frac{\text{Revenue from Ads}}{\text{Ad Spend}}$
- Metcalfe's Law: $V \propto n^2$

3. Class 06: Logistics & Supply Chain

Fulfillment Models

- **Dropshipping:** No inventory held; supplier ships directly to customer. Low risk, lower margins.
- **3PL (Third-Party Logistics):** Outsourcing warehousing to specialists (e.g., Amazon FBA, Pathao Fulfillment).
- **4PL (Fourth-Party Logistics):** An integrator that manages multiple 3PLs (e.g., **Alibaba's Cainiao Network**).

Inventory & EOQ

The **Economic Order Quantity (EOQ)** minimizes total cost (Ordering + Holding costs).

$$EOQ = \sqrt{\frac{2DS}{H}}$$

- D: Annual Demand
- S: Ordering Cost (per order)
- H: Holding Cost (per unit per year)

Last-Mile Challenges (Bangladesh Context)

- **Address Quality:** Non-standardized/descriptive addresses.
- **Traffic:** Dhaka congestion increases costs and reduces driver capacity.

- **Disaggregation:** High cost of splitting bulk shipments into individual home deliveries (53% of total shipping cost).

💡 Quick Case Studies

- **SSL Commerz:** Leading BD payment aggregator using a REST API.
- **Cainiao:** Alibaba's "asset-light, data-heavy" 4PL model that orchestrates 3rd party carriers via APIs.
- **A/B Testing:** Controlled experiment between Version A (Control) and B (Variant) to improve Conversion Rate.